

Nama : Zahra Nur Rohmah

NIM : 245410101

UTS

SISTEM TERDISTRIBUSI DAN TERDESENTRALISASI

3. Dengan menggunakan Docker / Docker Compose, buatlah streaming replication di PostgreSQL yang bisa menjelaskan sinkronisasi. Tulislah langkah-langkah pengerjaannya dan buat penjelasan secukupnya.

Langkah 1 : Membuat

- Folder postgres-repl : Folder proyek utama
- Folder postgres-repl/primary : Konfigurasi primary server
- Folder postgres-repl/replica : Konfigurasi replica server
- File docker-coompose.yml : YAML docker untuk menjalankan primary & replica

```
PS C:\Users\Asus> mkdir postgres-repl

Directory: C:\Users\Asus

Mode                LastWriteTime         Length Name
----                -
d-----          11/17/2025  11:07 AM                postgres-repl

PS C:\Users\Asus> cd postgres-repl
PS C:\Users\Asus\postgres-repl> mkdir primary

Directory: C:\Users\Asus\postgres-repl

Mode                LastWriteTime         Length Name
----                -
d-----          11/17/2025  11:08 AM                primary

PS C:\Users\Asus\postgres-repl> mkdir replica

Directory: C:\Users\Asus\postgres-repl

Mode                LastWriteTime         Length Name
----                -
d-----          11/17/2025  11:08 AM                replica

PS C:\Users\Asus\postgres-repl> notepad docker-coompose.yml
PS C:\Users\Asus\postgres-repl> docker compose up -d
no configuration file provided: not found
PS C:\Users\Asus\postgres-repl> docker compose up -d
no configuration file provided: not found
PS C:\Users\Asus\postgres-repl> dir *.y*

Directory: C:\Users\Asus\postgres-repl

Mode                LastWriteTime         Length Name
----                -
-a-----          11/17/2025  11:08 AM           572 docker-coompose.yml
```

Langkah 2 :

- Cek versi Docker Compose
- Membuat folder proyek
- Menjalankan Docker Compose
- Mengecek container
- Masuk ke container PRIMARY
- Masuk ke container REPLICA

```

C:\Users\Asus>docker compose version
Docker Compose version v2.40.3-desktop.1

C:\Users\Asus>postgres-replication/
'postgres-replication' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Asus>mkdir postgres-replication

C:\Users\Asus>cd postgres-replication

C:\Users\Asus\postgres-replication>notepad docker-compose.yml

C:\Users\Asus\postgres-replication>docker-compose up -d
time="2025-11-17T10:47:26+07:00" level=warning msg="C:\\Users\\Asus\\postgres-replication\\docker-comp
ose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion"
[+] Running 16/16
  ✓ primary Pulled                                         50.1s
  ✓ replica Pulled                                       49.5s
  ✓ d4b384052df7 Pull complete                          22.3s
  ✓ 4dc908580fc7 Pull complete                          7.2s
  ✓ 78c29b10a6ea Pull complete                         43.1s
  ✓ 829b70172da7 Pull complete                         21.1s
  ✓ 83e930c50688 Pull complete                          7.9s
  ✓ 8438d895aa42 Pull complete                          7.9s
  ✓ 60acc70d7b69 Pull complete                          7.9s
  ✓ eada273bc1ff Pull complete                         22.5s
  ✓ f8780632cce8 Pull complete                          7.9s
  ✓ d7ecdcd7702a Pull complete                         21.0s
  ✓ c535265a0c2c Pull complete                         21.6s
  ✓ 73c8530669d9 Pull complete                         21.5s
  ✓ 5d3e627a217e Pull complete                         43.0s
  ✓ 891b5a61c527 Pull complete                          7.9s
[+] Running 3/3
  ✓ Network postgres-replication_default Created         0.2s
  ✓ Container pg-primary Started                       2.4s
  ✓ Container pg-replica Started                       1.9s

C:\Users\Asus\postgres-replication>docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
5d5ccb11f7cb   postgres:15 "docker-entrypoint.s..." 31 seconds ago Up 29 seconds 0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
pg-primary

C:\Users\Asus\postgres-replication>docker exec -it pg-primary psql -U postgres
psql (15.15 (Debian 15.15-1.pgdg13+1))
Type "help" for help.

postgres=# CREATE TABLE test (id SERIAL PRIMARY KEY, name TEXT);
INSERT INTO test (name) VALUES ('Data dari PRIMARY');
CREATE TABLE
INSERT 0 1
postgres=# docker exec -it pg-replica psql -U postgres

```

Langkah 3: Membuat tabel di PRIMARY dan memasukkan ke REPLICA

```

postgres=# CREATE TABLE test (id SERIAL PRIMARY KEY, name TEXT);
INSERT INTO test (name) VALUES ('Data dari PRIMARY');
CREATE TABLE
INSERT 0 1
postgres=# docker exec -it pg-replica psql -U postgres
postgres=# SELECT * FROM test;

```

Langkah 4:

```

C:\Users\Asus\postgres-replication>docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
0cff7b51414b   postgres:15 "docker-entrypoint.s..." 5 minutes ago  Exited (1) 5 minutes ago
pg-replica
5d5ccb11f7cb   postgres:15 "docker-entrypoint.s..." 5 minutes ago  Up 5 minutes  0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
pg-primary

```

Langkah 5:

Container sudah berjalan dengan normal

```
C:\Users\Asus\postgres-replication>docker compose up -d
time="2025-11-17T10:58:36+07:00" level=warning msg="C:\\Users\\Asus\\postgres-replication\\docker-comp
ose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential
confusion"
[+] Running 2/2
  ✓ Container pg-primary   Running      0.0s
  ✓ Container pg-replica   Started      1.0s

C:\Users\Asus\postgres-replication>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
5d5ccb11f7cb   postgres:15   "docker-entrypoint.s..." 10 minutes ago Up 10 minutes  0.0.0.0:5432->5
432/tcp, [::]:5432->5432/tcp   pg-primary

C:\Users\Asus\postgres-replication>docker exec -it pg-primary psql -U postgres
psql (15.15 (Debian 15.15-1.pgdg13+1))
Type "help" for help.

postgres=# CREATE TABLE mahasiswa (
        id serial PRIMARY KEY,
        nama text
    );
CREATE TABLE
postgres=# INSERT INTO mahasiswa (nama) VALUES ('Budi'), ('Siti');
INSERT 0 2
```

Langkah 6: Cek Data di REPLICA

```
CREATE TABLE
postgres=# INSERT INTO mahasiswa (nama) VALUES ('Budi'), ('Siti');
INSERT 0 2
postgres=# SELECT * FROM mahasiswa;
 id | nama
----+-----
  1 | Budi
  2 | Siti
(2 rows)
```

```
services:
  primary:
    image: postgres:15
    container_name: pg-primary
    environment:
      POSTGRES_PASSWORD: postgres
    ports:
      - "5433:5432"
    volumes:
      - ./primary:/var/lib/postgresql/data
    networks:
      - pgnet

  replica:
    image: postgres:15
    container_name: pg-replica
    environment:
      POSTGRES_PASSWORD: postgres
    depends_on:
      - primary
    ports:
      - "5434:5432"
    volumes:
      - ./replica:/var/lib/postgresql/data
    networks:
      - pgnet

networks:
  pgnet:
```